

CHAPTER 1 RESULTS

1.1 Cohort Characteristics

TCGA-LUSC datasets for five modalities were retrieved (described in Section ??).

Gene expression and Copy number variation (CNV) retained the full labelled 178-sample cohort, whereas Deoxyribonucleic acid (DNA) methylation and miRNA expression had smaller assay overlap (Table 1.1). Across modalities, the classical subtype was the most frequent class and primitive the least frequent, motivating the use of macro-averaged classification metrics in subsequent analyses.

Table 1.1: Summary of omics modalities used in this study. Variation in sample size reflects assay availability and label overlap.

Modality	Configuration	Classical	Basal	Secretory	Primitive	Total
Gene expression	HiSeqV2, RNASeqV2 RSEM-normalised	65	43	43	27	178
DNA methylation	HumanMethylation450k beta values	29	14	22	11	76
miRNA expression	miRNA sequencing, log2(RPM + 1)	44	29	20	14	107
RPPA	Normalised protein abundance	38	28	28	18	112
CNV	GISTIC2 all-thresholded calls	65	43	43	27	178

1.2 Preprocessing Summary

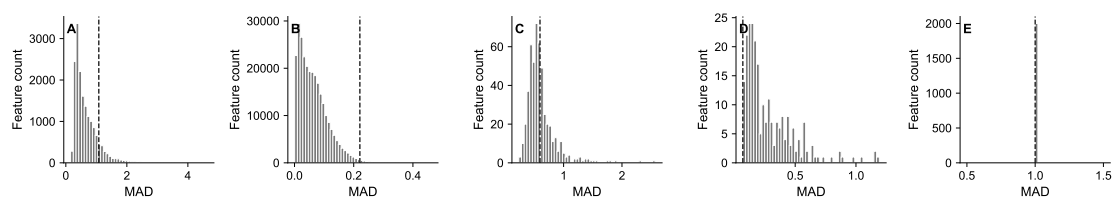


Figure 1.1: Feature variability distribution across processed omics modalities. Panels show MAD distributions for (A) gene expression, (B) DNA methylation, (C) micro RNA (miRNA) expression, (D) RPPA, and (E) CNV. Dashed vertical lines denote retained-feature threshold used for the processed matrix; for RPPA, all 238 antibodies were retained and the line denotes the minimum retained MAD.

Table 1.2: Preprocessing and feature-retention summary for each omics modality. Raw feature counts correspond to the input assay feature spaces. The MAD threshold is the score of the last retained feature in the processed matrix; for RPPA, where all 238 antibodies were retained, this value represents the minimum observed MAD among retained antibodies.

Modality	Raw features	Retained features	MAD threshold
Gene expression	60,660	2,000	1.0808
DNA methylation	485,577	2,000	0.2208
miRNA expression	1,881	200	0.5954
RPPA	238	238	0.0692
CNV	24,776	2,000	1.0000

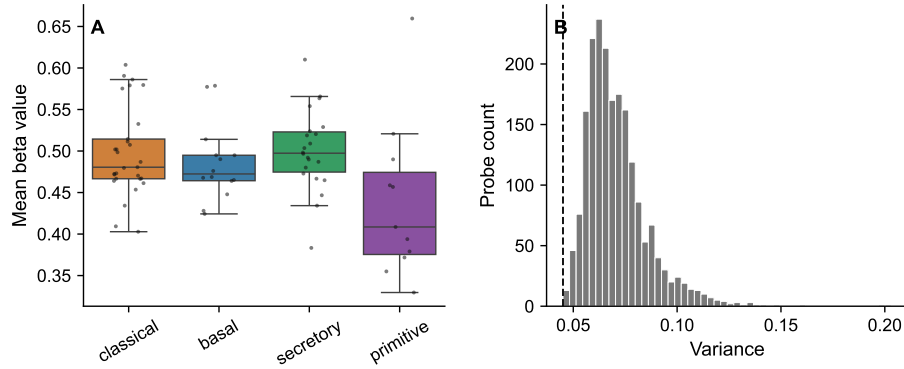


Figure 1.2: DNA methylation overview for labelled samples. (A) Mean beta value per sample stratified by subtype. Boxplots show median and interquartile range; points show individual samples. (B) Variance distribution across retained methylation probes. Dashed line denotes the minimum variance among retained probes in the processed matrix.

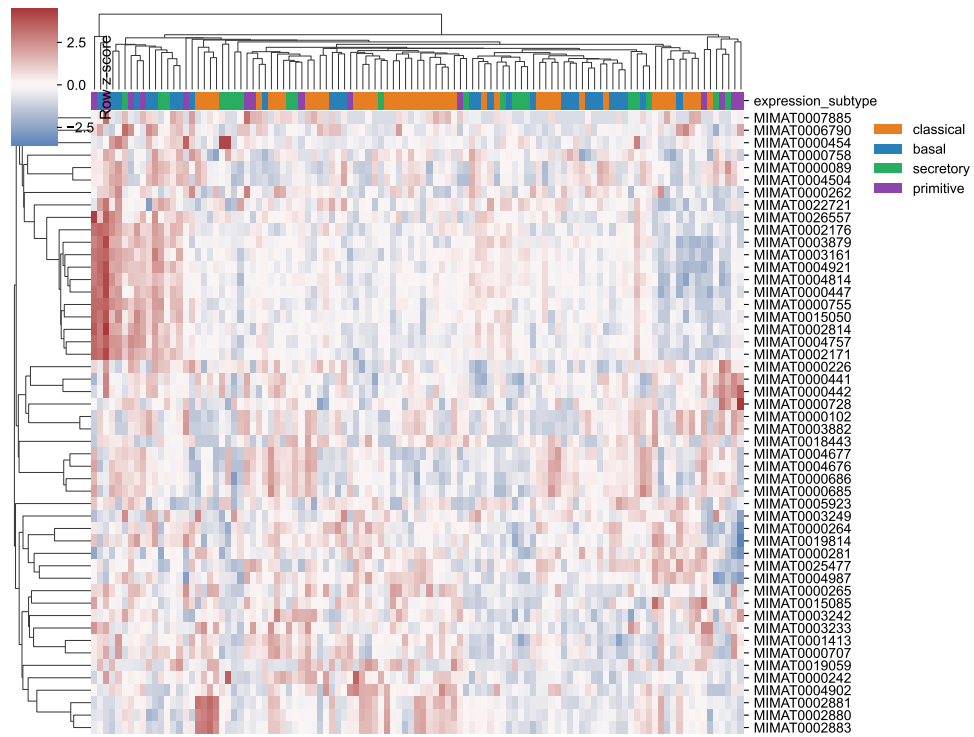


Figure 1.3: Clustered heatmap of 50 most variable miRNA features across labelled The Cancer Genome Atlas (TCGA)-Lung squamous cell carcinoma (LUSC) samples. Rows are miRNA features and columns are samples; values are row-scaled z-scores.

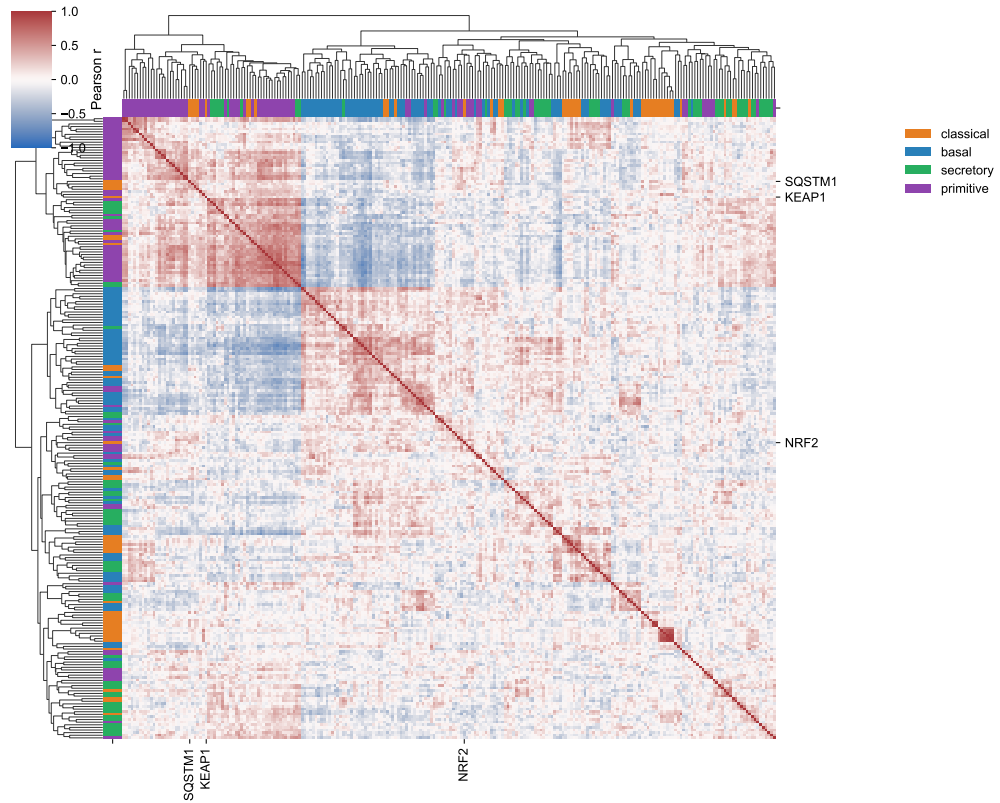


Figure 1.4: Hierarchically-clustered correlation heatmap across all 238 RPPA antibodies. Rows and columns represent antibodies, and colour denotes antibody-antibody correlation across RPPA samples. The top and side annotation strips indicate the expression subtype in which each antibody had the highest mean abundance.

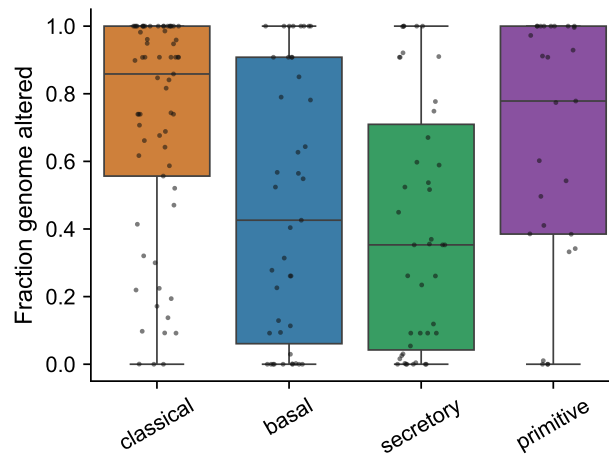


Figure 1.5: Fraction of retained CNV features with non-zero GISTIC2-thresholded copy-number state per sample, stratified by subtype. Boxplots show median and interquartile range; points show individual samples. Because the calculation is based on the 2,000 retained CNV features, it represents relative alteration burden within the processed CNV matrix.

1.3 Characterization of LUSC Subtypes

1.3.1 Clinical Characteristics

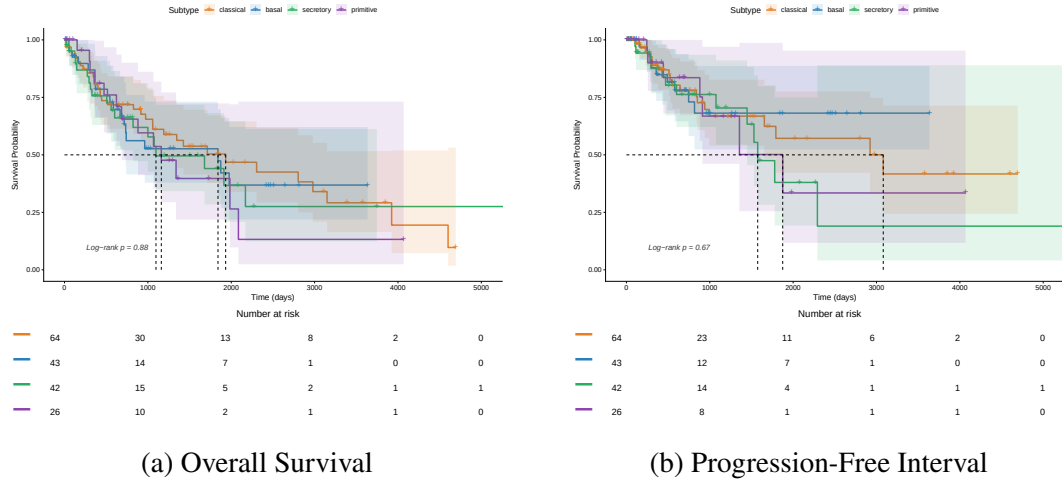


Figure 1.6: Survival curves showing Overall Survival (a) and Progression-Free Interval (b) for TCGA-LUSC samples (n=178) stratified by molecular subtypes. Median survival and 95% confidence intervals were computed by the Brookmeyer-Crowley method. Shaded regions denote 95% confidence intervals and risk table shows at-risk patient counts at selected time points.

Secretory and primitive subtype show the shortest median OS of 36 months and 38.1 months respectively, consistent with known proliferative and poorly differentiated biology of primitive subtypes (Hammerman et al., 2012; Wilkerson et al., 2010).

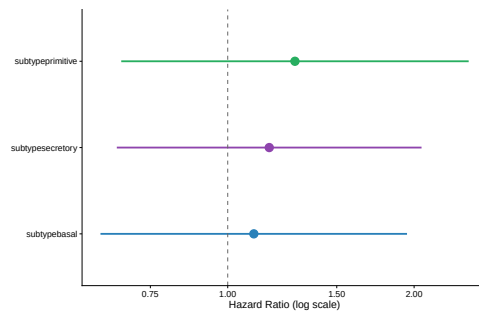


Figure 1.7: Hazard ratios for overall survival by LUSC molecular subtype from Cox proportional hazards regression, with classical subtype as reference (n=AVOCADO). Points denote hazard ratios and horizontal bars represent 95% confidence intervals. HR > 1 indicates worse prognosis relative to classical subtype. The proportional hazards assumption was verified using scaled Schoenfeld residuals (cox.zph global p=AVOCADO). Concordance index=AVOCADO

1.3.2 Immune Microenvironment

Mean EPIC cell fractions across LUSC expression subtypes

Top annotation shows BH-adjusted Kruskal-Wallis significance: * $q < 0.05$, ** $q < 0.01$, *** $q < 0.001$

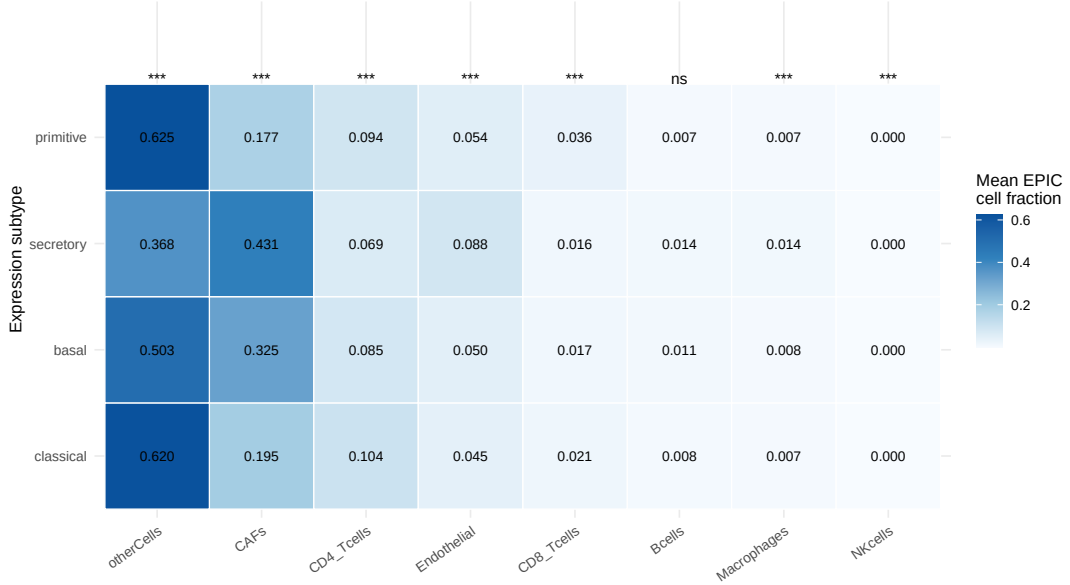


Figure 1.8: Mean EPIC cell fractions across LUSC subtypes ($n = 178$). Rows represent expression subtypes; columns represent EPIC-inferred cell compartments ordered by mean fraction. Numeric values show the mean fraction per subtype-cell-type combination. Significance annotations denote BH-adjusted Kruskal-Wallis q -values across subtypes (* $q < 0.05$, ** $q < 0.01$, *** $q < 0.001$; ns: not significant).

Secretory subtype showed the highest EPIC fractions for macrophages and CAFs; primitive subtype showed the highest fraction of uncharacterised cells (otherCells), consistent with its highly proliferative phenotype. B cells did not differ significantly across subtypes.

EPIC immune cell fractions by LUSC expression subtype

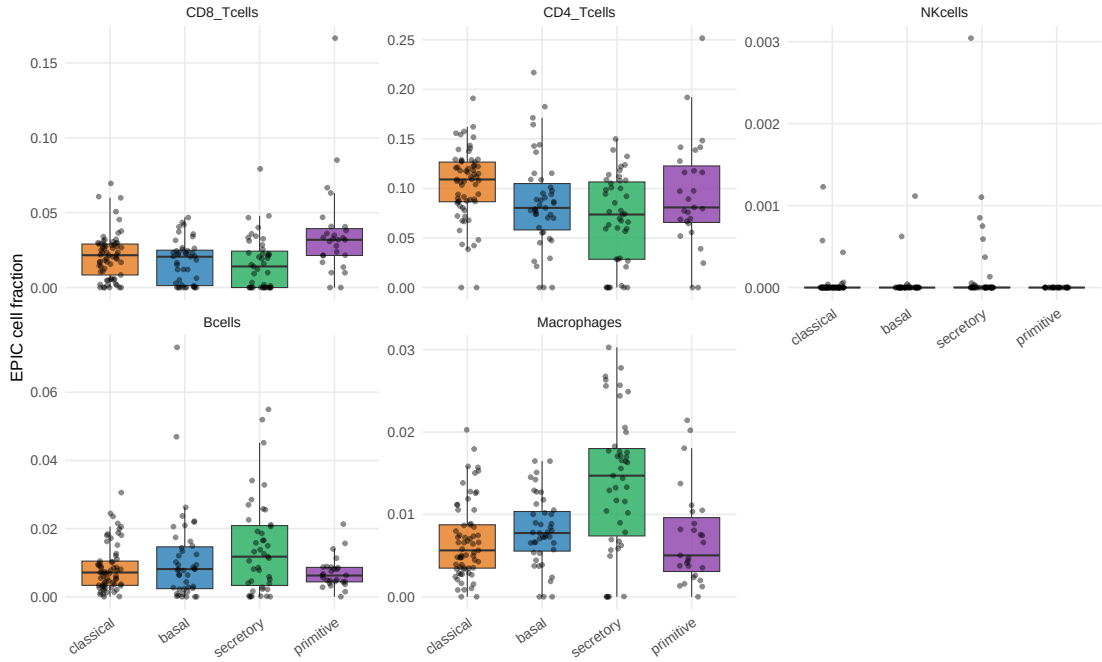


Figure 1.9: EPIC-estimated fractions for five cell populations across subtypes. Boxplots show median and interquartile range; individual points are samples.

xCell validation: immune enrichment scores by LUSC expression subtype

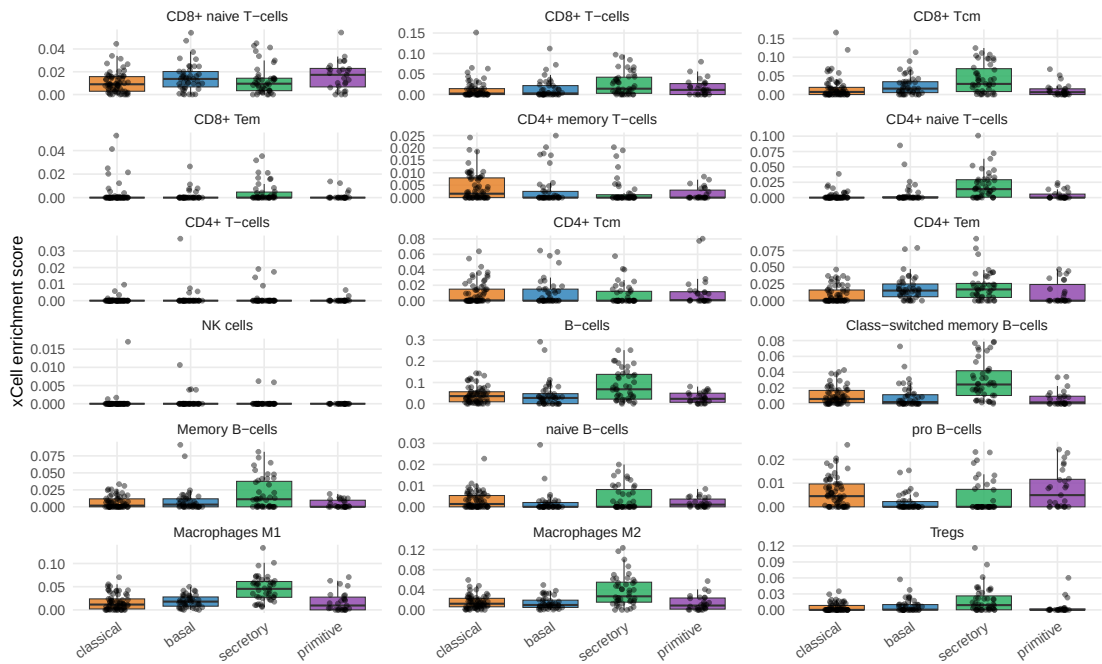


Figure 1.10: xCell enrichment scores for selected immune populations across subtypes. xCell scores are relative enrichment estimates, not cell fractions; scores across cell types are not directly comparable.

1.3.3 Somatic Mutation Landscape

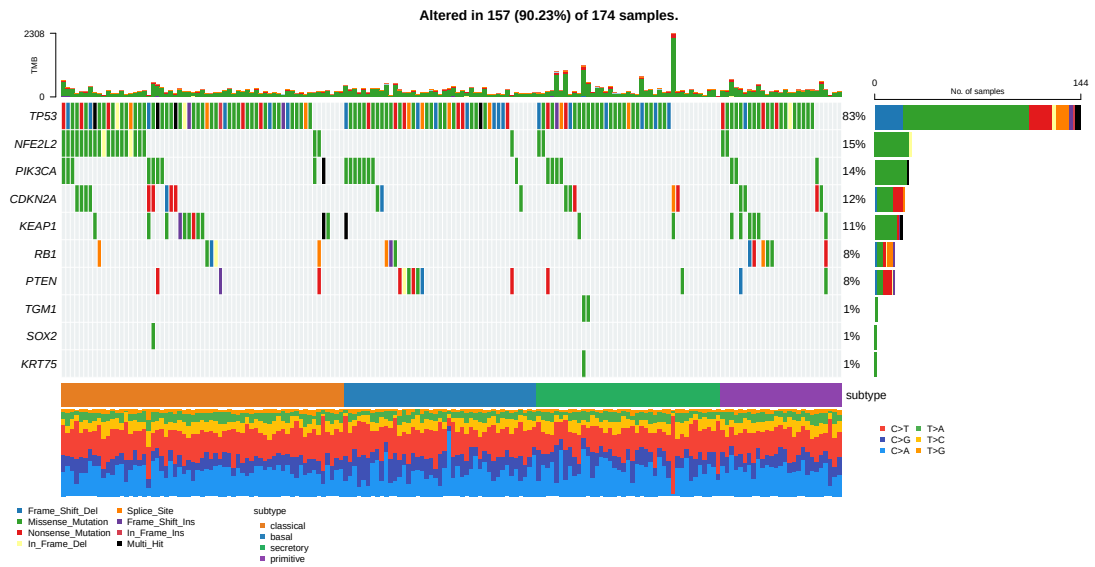


Figure 1.11: Oncoplot showing gene alterations across subtypes. Curated driver panel consisted of TP53, NFE2L2, KEAP1, RB1, PTEN, CDKN2A, PIK3CA, SOX2, FGFR1, TGM1 and KRT75.

TP53 was the most frequently mutated gene overall (Figure 1.11) but did not show subtype-specific enrichment, consistent with its role as a common subtype-independent driver alteration in LUSC.

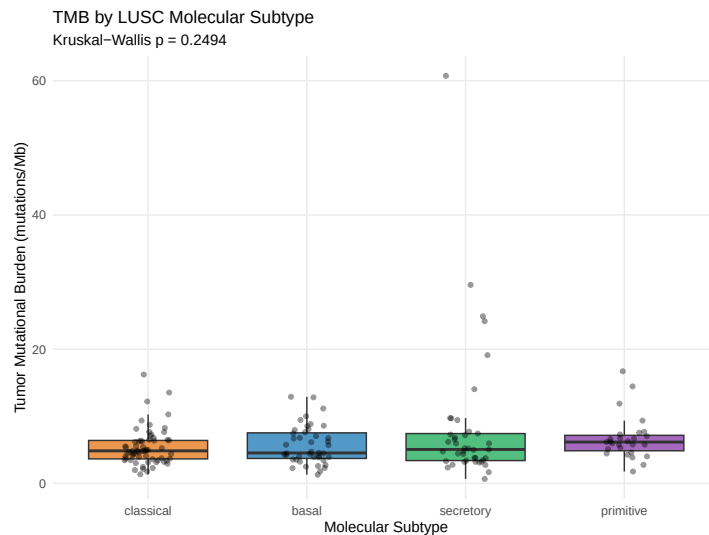


Figure 1.12: Tumour mutational burden by subtype.

TMB did not vary significantly across subtypes (Figure 1.12).

1.4 Dimensionality Structure and Subtype Separability

PCA-based separability analysis revealed substantial differences in subtype structure across omics modalities (Table 1.3).

1.4.1 Dimensionality Structure

Table 1.3: **AVOCADO**. PCs denotes the number of components required to represent 80% of the variance. S denotes Silhouette score. pseudo-F and p denote Permutational Multivariate Analysis of Variance (PERMANOVA) pseudo-F statistic and permutation p-value respectively.

Modality	n	Features	PCs	PC1(%)	S	pseudo-F	p
Gene expression	178	2000	62	12.3	0.047	8.997	0.001
miRNA expression	107	200	30	16.6	0.031	5.191	0.001
RPPA	112	238	27	16.4	0.008	3.935	0.001
DNA methylation	76	2000	11	57.7	-0.060	3.491	0.005
CNV	178	2000	5	44.7	-0.031	6.759	0.001

The number of principal components required to explain 80% of cumulative variance varied across platforms, reflecting differences in feature covariance structure. Gene expression (n=178, 2000 genes) required 62 components, with PC1 explaining 12.3% of variance. miRNA (n=107, 200 features) required 30 components and RPPA (n=112, 238 proteins) required 27 components. In contrast, methylation (n=76, 2000 probes) reached 80% variance in only 11 components, with PC1 explaining 57.7% variance. CNV (n=178, 2000 genes) was the most compact of all modalities, requiring only 5 components with PC1 explaining 44.7% and PC2 a further 20.8%.

1.4.2 Subtype Separability

PERMANOVA. All modalities showed statistically significant subtype group separation ($p \leq 0.005$), indicating that subtype centroids are non-random across every omics platform. Gene expression showed the strongest centroid separation (pseudo-F = 8.997, $p \leq 0.001$), followed by CNV (pseudo-F = 6.759, $p \leq 0.001$), miRNA (pseudo-F = 5.191, $p \leq 0.001$), RPPA (pseudo-F = 3.935, $p \leq 0.001$), and methylation (pseudo-F = 3.491, $p = 0.005$).

Silhouette scores. Silhouette scores showed remarkable discordance from PERMANOVA, with gene expression achieving the highest individual-sample cluster cohesion ($S = 0.047$), followed by miRNA ($S = 0.031$), RPPA ($S = 0.008$), CNV ($S = -0.031$), and methylation ($S = -0.060$). Negative silhouette scores for CNV and methylation indicate

that, on average, samples are geometrically closer to an adjacent subtype's centroid than their own, reflecting high within-group variance relative to between-group separation, even when centroids are statistically distinguishable.

Per-subtype Silhouette scores. Primitive subtype showed negative mean silhouette scores across all modalities (GE: -0.062; miRNA: -0.133; RPPA: -0.047; methylation: -0.011; CNV: -0.224), indicating that primitive samples are consistently closer to other subtype centroids than to their own. Classical and basal subtypes showed highest silhouette scores across modalities, suggesting more coherent cluster membership in their respective omics feature spaces.

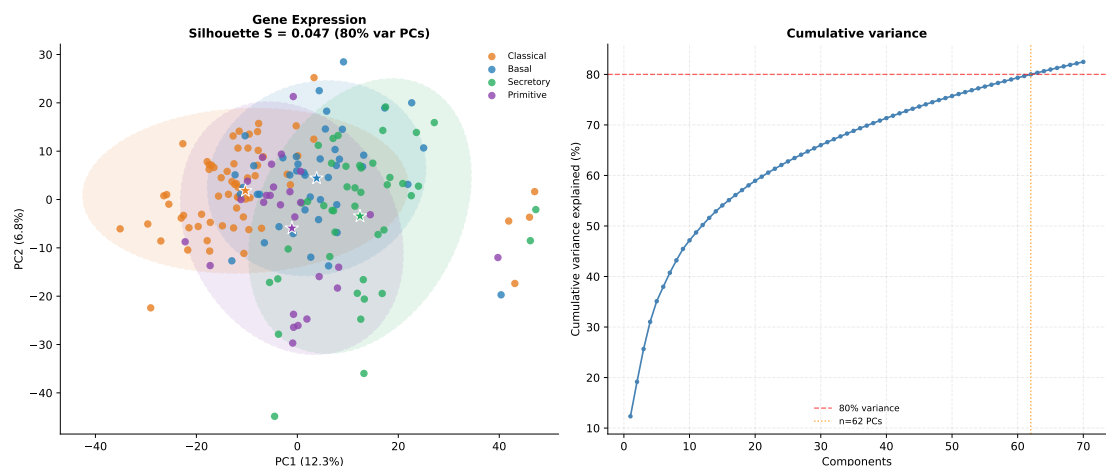


Figure 1.13: Left figure shows PC1 versus PC2 for gene expression; variance explained by each component is shown on the axis label.

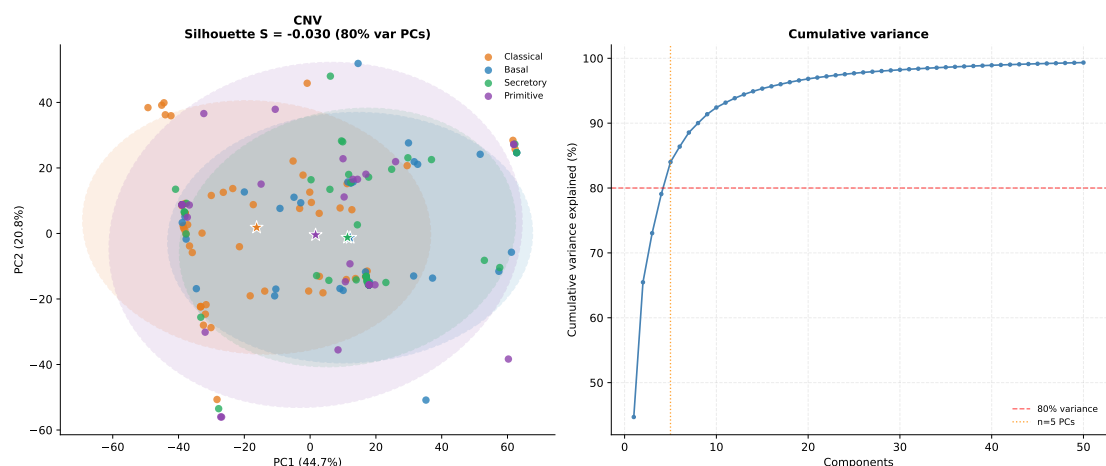


Figure 1.14: Left figure shows PC1 versus PC2 for CNV; variance explained by each component is shown on the axis label.

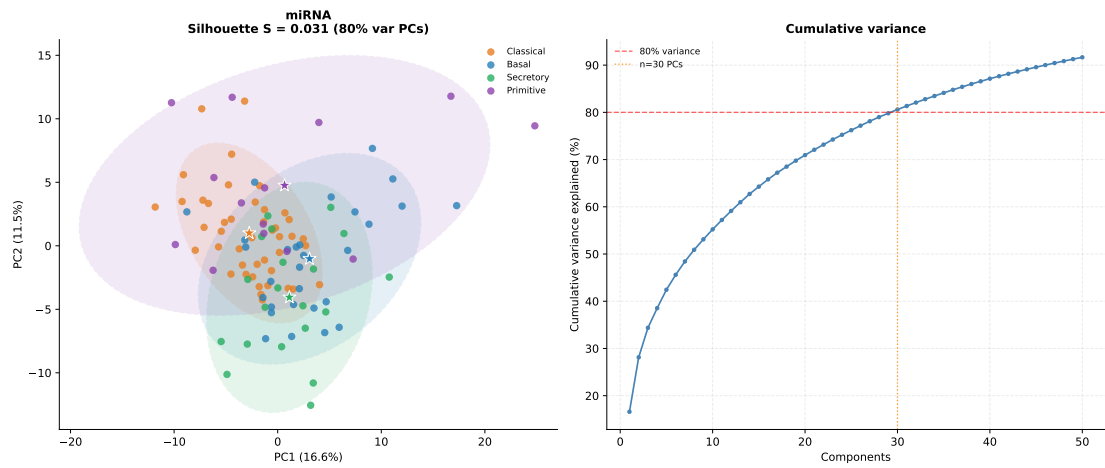


Figure 1.15: Left figure shows PC1 versus PC2 for miRNA; variance explained by each component is shown on the axis label.

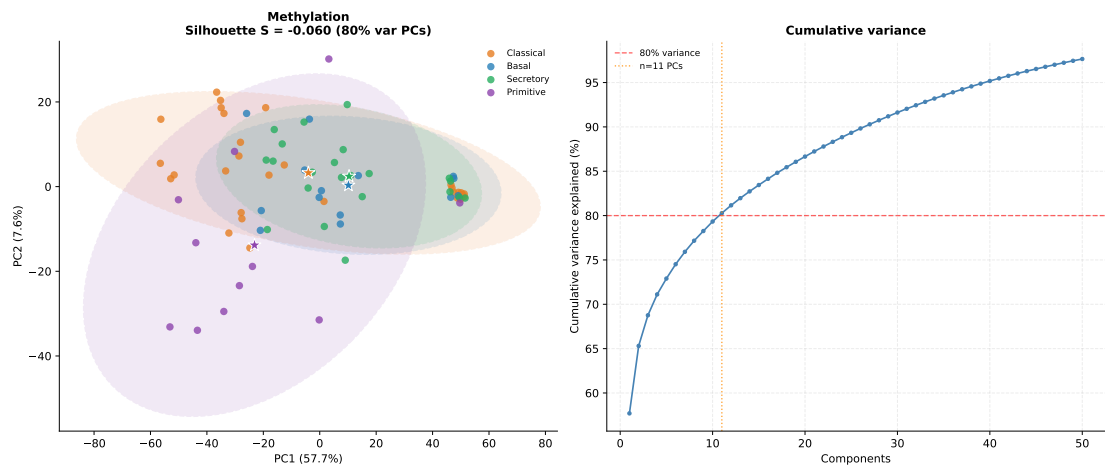


Figure 1.16: Left figure shows PC1 versus PC2 for methylation; variance explained by each component is shown on the axis label.

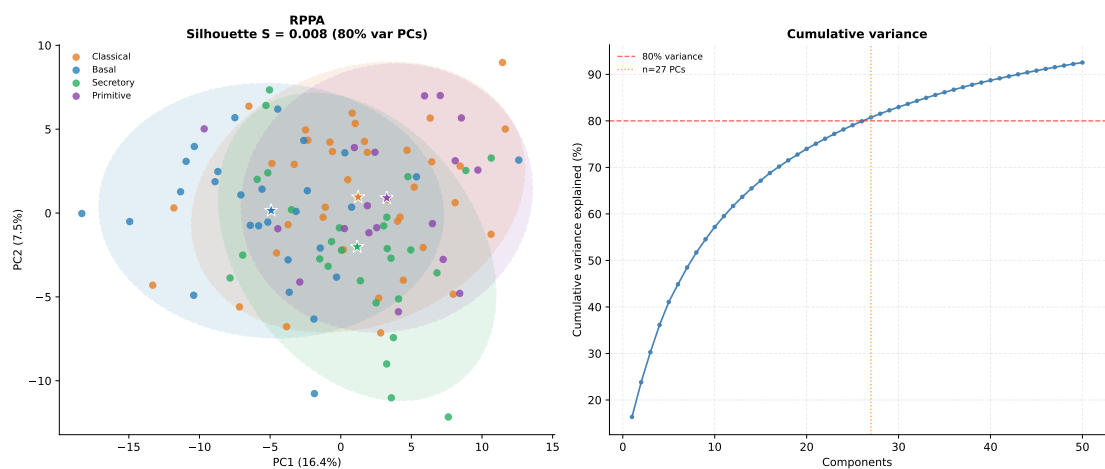


Figure 1.17: Left figure shows PC1 versus PC2 for RPPA; variance explained by each component is shown on the axis label.

- 1.5 Performance Across Omics Modalities**
- 1.6 SHAP Feature Importance and Stability**
- 1.7 Cross-Model Feature Overlap**
- 1.8 Cross-Modality Feature Overlap**
- 1.9 Pathway Enrichment**

References

- Hammerman, P. S., Lawrence, M. S., Voet, D., Jing, R., Cibulskis, K., Sivachenko, A., Stojanov, P., McKenna, A., Lander, E. S., Gabriel, S., Getz, G., Sougnez, C., Imielinski, M., Helman, E., Hernandez, B., Pho, N. H., Meyerson, M., Chu, A., Chun, H.-J. E., ... group, D. w. (2012). Comprehensive genomic characterization of squamous cell lung cancers. *Nature*, 489(7417), 519–525. <https://doi.org/10.1038/nature11404>
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